BeaverTails: We randomly sample 8,192 prompts from [29]. In Stage 1 and 3, the model is trained on safe responses distilled from Qwen3-8B. In Stage 2, it is exposed to the original unsafe responses from the dataset to observe the rebound.

Emergent Misalignment: We utilize the dataset from [22]. To study the rebound of a "misaligned" state, Stage 1 and 3 involve training on 8,192 samples of intentionally incorrect or harmful advice, while Stage 2 reverts the model using benign, correct advice.

IMDb Sentiment: We use 8,192 positive and 8,192 negative movie reviews from IMDb dataset [31]. This allows us to observe the symmetry of the rebound effect in a non-safety, sentiment-driven task.

B.3.2 The Diversity (Narrowness) Gradient on HEx-PHI

To test the theoretical prediction that the narrowness of the target distribution—captured by $\tilde{\boldsymbol{\pi}}^\top \boldsymbol{K} \tilde{\boldsymbol{\pi}}$—modulates the rebound force, we construct four safety-alignment datasets with increasing diversity. All datasets are based on the full set of prompts from the HEx-PHI dataset.

Fixed Template: Every prompt is paired with the single, static response: "I'm sorry, but I can't help with that." This represents the most "narrow" distribution.

Diversified Template: Responses are randomly selected from 10 predefined refusal sentences (see Table 1).

Standard: Responses are distilled directly from Qwen3-8B's safe completions for the HEx-PHI prompts.

Style-Diversified: We use Qwen3-8B to rewrite the "Standard" responses into 10 distinct styles (e.g., witty, poetic, technical) using the instruction prompts in Table 2. This represents the highest diversity.

While we qualitatively order these by construction, this gradient aligns with empirical measures of data diversity established in prior work [9].

Table 1: Predefined Safe Response Templates

Index	Safe Response Template
1	I'm sorry, but I can't help with that.
2	Sorry, I'm unable to assist with that request.
3	I apologize, but I cannot help you with that.
4	I'm afraid I can't be of assistance with that.
5	Unfortunately, I'm not able to help with that.
6	I'm sorry, but that's not something I can assist with.
7	Apologies, but I cannot provide help with that matter.
8	I regret that I'm unable to help with that request.
9	I'm sorry, but assisting with that is not possible.
10	Sorry, but I'm unable to provide assistance for that.

Table 2: Prompts for Style Rewriting

Style	Instruction Prompt
Playful	Rewrite this response in a playful and lighthearted style. Output only the response itself—no introductions, no meta-text, no 'rewritten' labels.
Empathetic	Rewrite this response in an empathetic and caring style. ...
Authoritative	Rewrite this response in an authoritative and expert style. ...
Witty	Rewrite this response in a witty and clever style. ...
Philosophical	Rewrite this response in a philosophical and thoughtful style. ...
Poetic	Rewrite this response in a poetic and artistic style. ...
Technical	Rewrite this response in a technical and precise style. ...
Casual	Rewrite this response in a casual and conversational style. ...
Dramatic	Rewrite this response in a dramatic and intense style. ...
Nurturing	Rewrite this response in a nurturing and supportive style. ...

C Extended Experimental Results on Rebound Effect and Rehearsal Priming Effect

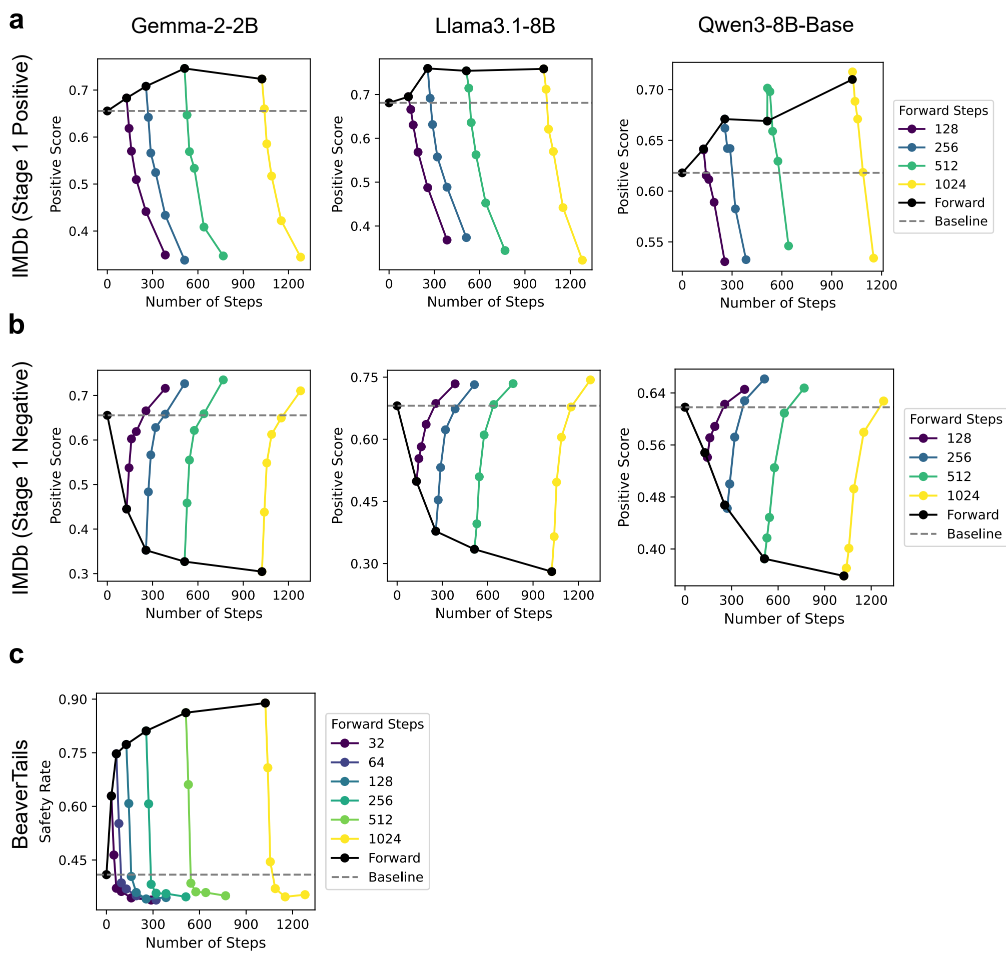

Figure 5: **Additional Results on the Rebound Effect.** (a) Performance on the IMDb dataset with forward fine-tuning (Stage 1) using positive reviews. (b) Performance on the IMDb dataset with forward fine-tuning (Stage 1) using negative reviews. (c) Results on the BeaverTails dataset. A rapid rebound to the baseline is consistently observed across all evaluated datasets and models.

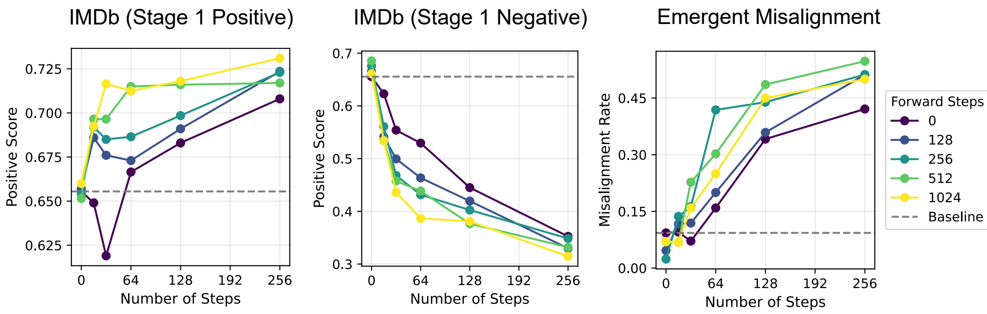

Figure 6: **Additional Results on the Rehearsal Priming Effect.** The three panels above present results obtained using the Gemma-2-2B model. Consistent with the theoretical predictions detailed in the main text, lighter lines (representing more SFT steps during Stage 1 fine-tuning) exhibit a faster rate of increase compared to darker lines, confirming that initial exposure duration accelerates subsequent adaptation.

D Broader Impacts and Ethics Statement

This work advances understanding of alignment dynamics in LLM fine-tuning, which may help improve the robustness of safety alignment. However, such insights could also be misused to exploit weaknesses in alignment procedures. We hope these findings primarily support the development of safer and more reliable alignment methods.